WORKSHEET 8.1: DEPTH I

Throughout this course, "ring" means *commutative* ring with unity and all ring homomorphisms preserve unity. In the previous worksheets we introduced Krull dimension and height, and saw that finitely generated algebras over a field can often be understood by cutting them down to polynomial rings. The goal of this worksheet is to study sequences of elements that behave like independent equations. The algebraic word for this independence is *regularity*.

A regular sequence is a list of equations such that each new equation remains a nonzerodivisor after imposing the previous ones. This condition is strong enough to force dimension-theoretic consequences: by Krull's Principal Ideal Theorem, one equation cuts height by at most one, and a regular equation really does cut height by exactly one. The same idea leads to *depth*, which measures how long a regular sequence can be found inside the maximal ideal of a local ring.

The final part of the worksheet introduces the *Koszul complex*. This is a small, explicit chain complex attached to a sequence of ring elements. When the sequence is regular, the Koszul complex is exact except in degree zero. Thus the Koszul complex packages regular sequences, depth computations, and explicit syzygies into one object.

Let R be a ring and let M be an R-module. An element $x \in R$ is a *nonzerodivisor on M* if the multiplication map

$$M \longrightarrow M, \qquad m \longmapsto xm$$

is injective. We say that x is *M-regular* if x is a nonzerodivisor on M and $M/xM \neq 0$. A sequence $x_1, \dots, x_r \in R$ is an *M-regular sequence* if x_1 is M-regular and, for each $i \geq 2$, the image of x_i is regular on

$$M/\langle x_1, \dots, x_{i-1}\rangle M.$$

When $M = R$, we simply say that $x_1, \dots, x_r$ is an *R-regular sequence*, or just a *regular sequence* if the ring is clear.

---

(1) **First Encounters with Regular Elements.** Let M be an R-module and let $x \in R$.

(a) Show that x is a nonzerodivisor on M if and only if

$$(0 :_M x) := \{m \in M \mid xm = 0\} = 0.$$

(b) Show that if R is a domain, then every nonzero element of R is a nonzerodivisor on R. Which of these elements are R-regular?

(c) Let $(R, \mathfrak{m})$ be a local ring and let M be a finitely generated nonzero R-module. If $x \in \mathfrak{m}$ is a nonzerodivisor on M, prove that $M/xM \neq 0$. Hint: Use Nakayama's lemma.

© Juliette Bruce 2026.
Licensed under Creative Commons BY-NC-SA 4.0.
Inspired by © Karen Smith (2019); see.

(d) Let $R = \mathbb{K}[x,y]$. Prove that x, y, and $x+y$ are nonzerodivisors on R.

(e) Let $R = \mathbb{K}[x,y]/\langle xy\rangle$. Show that the images of x and y are zerodivisors on R, but the image of $x+y$ is a nonzerodivisor. Hint: Write every element uniquely as

$$c + xf(x) + yg(y) \qquad \text{with } c \in \mathbb{K},\ f \in \mathbb{K}[x],\ g \in \mathbb{K}[y].$$

(f) Let $R = \mathbb{K}[x,y]/\langle x^2, xy\rangle$. Show that the image of x is killed by the maximal ideal $\langle x, y\rangle$.

(2) **Building Regular Sequences.**

(a) Prove that $x_1,\ldots,x_r$ is a regular sequence on $\mathbb{K}[x_1,\ldots,x_n]$ for every $r \le n$.

(b) More generally, let A be a nonzero ring. Prove that $t_1,\ldots,t_r$ is a regular sequence on $A[t_1,\ldots,t_n]$ for every $r \le n$.

(c) In $\mathbb{K}[x,y]$, show that x, y is a regular sequence but x, xy is not a regular sequence.

(d) In $\mathbb{K}[x,y]/\langle xy\rangle$, show that $x+y$ is a regular element but $x+y, x$ is not a regular sequence.

(e) Suppose $x_1,\ldots,x_r$ is an M-regular sequence. Prove that every initial segment $x_1,\ldots,x_i$ is an M-regular sequence.

(f) With notation as in the previous part, prove that $x_{i+1},\ldots,x_r$ is regular on $M/\langle x_1,\ldots,x_i\rangle M$.

(g) Let $U \subset R$ be multiplicatively closed. Prove that the image of an M-regular sequence $x_1,\ldots,x_r$ is a sequence of nonzerodivisors on $U^{-1}M$, after successively quotienting by the previous elements. What extra condition is needed to ensure that it is a regular sequence in the sense that the final quotients are nonzero?

(3) **Regular Sequences and Permutations.** In a general ring, the order of a regular sequence is part of the data. In a Noetherian local ring, however, regular sequences inside the maximal ideal can be permuted.

Theorem 1 (Permuting Regular Sequences)**.** *Let $(R,\mathfrak{m})$ be a Noetherian local ring and let M be a finitely generated R-module. If $x_1,\ldots,x_r \in \mathfrak{m}$ is an M-regular sequence, then every permutation of $x_1,\ldots,x_r$ is an M-regular sequence.*

(a) Use the theorem to prove that if x, y is a regular sequence in a Noetherian local ring, then y, x is a regular sequence.

(b) Let $R = \mathbb{K}[x,y]_{\langle x,y\rangle}$. Prove that x, y and y, x are both regular sequences.

(c) Let $R = \mathbb{K}[x,y]_{\langle x,y\rangle}/\langle xy\rangle$. Explain why x, y is not a regular sequence, even though both x and y lie in the maximal ideal.

(d) Let $R = \mathbb{K}[x,y,z]_{\langle x,y,z\rangle}$. Prove that $x, y+z, z$ is a regular sequence. Then use the theorem to produce five more regular sequences from the same three elements.

Regular sequences are meant to model independent equations. Krull's Principal Ideal Theorem is the first formal statement that one equation cannot cut out something of arbitrarily large codimension.

Theorem 2 (Krull's Principal Ideal Theorem)**.** *Let R be a Noetherian ring and let $x \in R$. If $\mathfrak{p} \in \operatorname{Spec}(R)$ is minimal over the principal ideal $\langle x \rangle$, then*

$$\operatorname{ht}(\mathfrak{p}) \leq 1.$$

More generally, if an ideal can be generated by r elements, then every prime minimal over that ideal has height at most r. This is often called *Krull's Height Theorem*. Thus the number of equations gives an upper bound on the height of every irreducible component they cut out. Regular sequences will give the matching lower bound.

---

(4) **Using the Principal Ideal Theorem.** Let R be a Noetherian ring.

(a) Let $x \in R$ and let $\mathfrak{p}$ be minimal over $\langle x \rangle$. Explain why localizing at $\mathfrak{p}$ reduces the Principal Ideal Theorem to the case of a local ring $(R, \mathfrak{m})$ in which $\mathfrak{m}$ is minimal over $\langle x \rangle$.

(b) Let $R = \mathbb{K}[x_1, \ldots, x_n]$ and let $0 \neq f \in R$. If $\mathfrak{p}$ is minimal over $\langle f \rangle$, use the Principal Ideal Theorem to show that $\operatorname{ht}(\mathfrak{p}) = 1$.

(c) Suppose x is a nonzerodivisor on R. Prove that no minimal prime of R contains x. Hint: If $\mathfrak{q}$ is minimal and $x \in \mathfrak{q}$, then $x/1$ is nilpotent in $R_{\mathfrak{q}}$.

(d) Deduce that if x is a nonzerodivisor and $\mathfrak{p}$ is minimal over $\langle x \rangle$, then $\operatorname{ht}(\mathfrak{p}) = 1$.

(e) Give an example showing that the nonzerodivisor hypothesis in the previous part is necessary. Hint: Use $\mathbb{K}[x, y]/\langle xy \rangle$ and the element x.

(5) **From Principal Ideals to Finitely Generated Ideals.** The goal of this exercise is to prove Krull's Height Theorem from the Principal Ideal Theorem. Let $I = \langle x_1, \ldots, x_r \rangle$ and let $\mathfrak{p}$ be minimal over I.

(a) Prove the theorem when $r = 1$.

(b) Assume $r > 1$. Choose a prime ideal $\mathfrak{q} \subseteq \mathfrak{p}$ minimal over $\langle x_1, \ldots, x_{r-1} \rangle$. Explain why such a $\mathfrak{q}$ exists.

(c) By induction, show that $\operatorname{ht}(\mathfrak{q}) \leq r - 1$.

(d) Show that $\mathfrak{p}/\mathfrak{q}$ is minimal over the principal ideal generated by the image of x_r in $R/\mathfrak{q}$.

(e) Apply the Principal Ideal Theorem in $R/\mathfrak{q}$ to show that the largest length of a chain of primes from $\mathfrak{q}$ to $\mathfrak{p}$ is at most 1.

(f) Conclude that $\operatorname{ht}(\mathfrak{p}) \leq r$.

(g) Use the theorem to compute the heights of the minimal primes over the following ideals:

(i) $\langle x, y\rangle \subset \mathbb{K}[x,y,z]$.

(ii) $\langle xy, z\rangle \subset \mathbb{K}[x,y,z]$.

(iii) $\langle xz, yz\rangle \subset \mathbb{K}[x,y,z]$.

(iv) $\langle x^2, xy\rangle \subset \mathbb{K}[x,y]$.

(6) **Regular Sequences Force Height.** Let $(R,\mathfrak{m})$ be a Noetherian local ring.

(a) Suppose $x \in \mathfrak{m}$ is a nonzerodivisor. Let

$$\mathfrak{p}_0/\langle x\rangle \subsetneq \mathfrak{p}_1/\langle x\rangle \subsetneq \cdots \subsetneq \mathfrak{p}_s/\langle x\rangle$$

be a chain of prime ideals in $R/\langle x\rangle$. Choose a minimal prime $\mathfrak{q}$ of R contained in $\mathfrak{p}_0$. Use the previous exercise to show $x \notin \mathfrak{q}$, and deduce that

$$\mathfrak{q} \subsetneq \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_s$$

is a chain in R.

(b) Deduce that if $x \in \mathfrak{m}$ is a nonzerodivisor, then

$$\dim(R/\langle x\rangle) \leq \dim(R) - 1.$$

(c) Use induction to prove that every regular sequence in $\mathfrak{m}$ has length at most $\dim(R)$.

(d) Now let R be any Noetherian ring and let $x_1,\ldots,x_r$ be an R-regular sequence. If $\mathfrak{p}$ is minimal over $I = \langle x_1,\ldots,x_r\rangle$, localize at $\mathfrak{p}$ and prove that $\mathrm{ht}(\mathfrak{p}) \geq r$.

(e) Combine the previous part with Krull's Height Theorem to prove that every prime minimal over I has height exactly r.

(7) **Complete Intersection Examples.** Let $R = \mathbb{K}[x,y,z]$.

(a) Show that x, z is a regular sequence. Compute the height of $\langle x, z\rangle$.

(b) Show that x, yz is a regular sequence. Find the minimal primes over $\langle x, yz\rangle$ and compute their heights.

(c) Show that xy, xz is not a regular sequence. Find a minimal prime over $\langle xy, xz\rangle$ of height 1.

(d) Explain how the previous parts illustrate the difference between being generated by two elements and being generated by a regular sequence of length two.

Let $(R,\mathfrak{m})$ be a Noetherian local ring and let M be a finitely generated nonzero R-module. The *depth* of M is

$$\mathrm{depth}_R(M) := \sup\{r \in \mathbb{Z}_{\geq 0} \mid \text{there exists an } M\text{-regular sequence } x_1,\ldots,x_r \in \mathfrak{m}\}.$$

Equivalently, depth measures how many independent equations from the maximal ideal can act nontrivially on M in succession. We write $\mathrm{depth}(R)$ for $\mathrm{depth}_R(R)$. One should think of depth as a measure of how far M is from having embedded infinitesimal pieces supported only at the closed point.

The dimension bound from the previous section implies

$$\operatorname{depth}_R(M) \leq \dim_R(M),$$

where $\dim_R(M)$ denotes the dimension of the support of M. In particular, for $M = R$ one has $\operatorname{depth}(R) \leq \dim(R)$. A Noetherian local ring is called *Cohen–Macaulay* if

$$\operatorname{depth}(R) = \dim(R).$$

Thus Cohen–Macaulay rings are rings in which the largest possible regular sequences actually exist.

---

(8) **First Depth Computations.** Let $(R, \mathfrak{m}, \Bbbk)$ be a Noetherian local ring.

(a) Prove that $\operatorname{depth}_R(\Bbbk) = 0$.

(b) If R is a domain and not a field, prove that $\operatorname{depth}(R) \geq 1$.

(c) Let $S = \Bbbk[x_1, \ldots, x_n]_{\langle x_1, \ldots, x_n \rangle}$. Prove that $\operatorname{depth}(S) = n$.

(d) For $0 \leq r \leq n$, compute

$$\operatorname{depth}_S(S/\langle x_1, \ldots, x_r \rangle).$$

(e) Let $A = \Bbbk[x, y]_{\langle x, y \rangle}/\langle xy \rangle$. Show that $\operatorname{depth}(A) = 1$. Hint: Use the image of $x + y$.

(f) Let $B = \Bbbk[x, y]_{\langle x, y \rangle}/\langle x^2, xy \rangle$. Show that $\operatorname{depth}(B) = 0$. Hint: Find a nonzero element killed by the maximal ideal.

(9) **Depth and Complete Intersections.** The following theorem is one of the basic reasons regular sequences are useful.

Theorem 3 (Depth Drops Along Regular Elements)**.** *Let $(R, \mathfrak{m})$ be a Noetherian local ring and let M be a finitely generated R-module. If $x \in \mathfrak{m}$ is M-regular, then*

$$\operatorname{depth}_R(M/xM) = \operatorname{depth}_R(M) - 1.$$

More generally, if $x_1, \ldots, x_r \in \mathfrak{m}$ is an M-regular sequence, then

$$\operatorname{depth}_R\big(M/\langle x_1, \ldots, x_r \rangle M\big) = \operatorname{depth}_R(M) - r.$$

(a) Use the theorem to compute the depth of $S/\langle f \rangle$, where $S = \Bbbk[x_1, \ldots, x_n]_{\langle x_1, \ldots, x_n \rangle}$ and $0 \neq f \in \langle x_1, \ldots, x_n \rangle$.

(b) Suppose $x_1, \ldots, x_r$ is a regular sequence in $S = \Bbbk[x_1, \ldots, x_n]_{\langle x_1, \ldots, x_n \rangle}$. Prove that $S/\langle x_1, \ldots, x_r \rangle$ is Cohen–Macaulay.

(c) Use this to prove that $\Bbbk[x, y]_{\langle x, y \rangle}/\langle xy \rangle$ is Cohen–Macaulay.

(d) Explain why $\Bbbk[x, y]_{\langle x, y \rangle}/\langle x^2, xy \rangle$ is not Cohen–Macaulay.

(e) Let $S = \Bbbk[x, y, z]_{\langle x, y, z \rangle}$. Show that $S/\langle x, yz \rangle$ is Cohen–Macaulay and compute its dimension.

(10) **Depth Zero.** Let $(R, \mathfrak{m})$ be Noetherian local and let M be a finitely generated nonzero R-module.

(a) Show that if there exists $0 \neq m \in M$ such that $\mathfrak{m}m = 0$, then $\operatorname{depth}_R(M) = 0$.

(b) The converse is also true, but requires the theorem that the zerodivisors on a finite module are the union of its associated primes. Assuming this theorem, prove that if $\operatorname{depth}_R(M) = 0$, then there exists $0 \neq m \in M$ such that $\mathfrak{m}m = 0$.

(c) Interpret the element $\overline{x} \in \mathbb{K}[x,y]_{\langle x,y\rangle}/\langle x^2, xy\rangle$ using part (a).

(d) Give an example of a local ring of depth zero and positive dimension.

We now introduce a complex that detects regular sequences. Start with one element $x \in R$. The *Koszul complex on x with coefficients in M* is

$$\operatorname{Kos}(x;M): \qquad 0 \longrightarrow M \xrightarrow{x} M \longrightarrow 0,$$

where the left copy of M is in homological degree 1 and the right copy is in degree 0. Its homology is easy to read:

$$H_1(\operatorname{Kos}(x;M)) = (0 :_M x), \qquad H_0(\operatorname{Kos}(x;M)) = M/xM.$$

Thus the one-element Koszul complex is exact in positive degree exactly when x is a nonzerodivisor on M.

For a sequence $\mathbf{x} = x_1, \ldots, x_n$, let $F = R^n$ with basis $e_1, \ldots, e_n$. The *Koszul complex* $\operatorname{Kos}(\mathbf{x};M)$ is the complex with

$$\operatorname{Kos}_i(\mathbf{x};M) = \bigwedge^i F \otimes_R M$$

and differential determined by

$$d(e_{j_1} \wedge \cdots \wedge e_{j_i} \otimes m) = \sum_{\ell=1}^{i} (-1)^{\ell-1} e_{j_1} \wedge \cdots \wedge \widehat{e_{j_\ell}} \wedge \cdots \wedge e_{j_i} \otimes x_{j_\ell} m.$$

One checks that $d^2 = 0$, so this is indeed a chain complex. We write $H_i(\mathbf{x};M)$ for its ith homology module.

---

(11) **The One-Element Koszul Complex.** Let $x \in R$.

(a) Compute $H_1(x;M)$ and $H_0(x;M)$ directly from the definition.

(b) Show that $H_1(x;M) = 0$ if and only if x is a nonzerodivisor on M.

(c) Let $R = \mathbb{Z}$ and $M = \mathbb{Z}$. Compute $H_i(n;\mathbb{Z})$ for $n \in \mathbb{Z}$.

(d) Let $R = \mathbb{K}[\epsilon]/\langle \epsilon^2 \rangle$. Compute $H_i(\epsilon;R)$.

(e) Let $R = \mathbb{K}[x]$ and $M = R/\langle x^m \rangle$. Compute $H_i(x;M)$.

(12) **The Two-Element Koszul Complex.** Let $x, y \in R$. The Koszul complex $\operatorname{Kos}(x,y;M)$ has the form

$$0 \longrightarrow M \xrightarrow{d_2} M^2 \xrightarrow{d_1} M \longrightarrow 0,$$

where

$$d_2(m) = (-ym, xm), \qquad d_1(a,b) = xa + yb.$$

(a) Verify that $d_1 d_2 = 0$.

(b) Prove that $H_0(x,y;M) \cong M/\langle x,y\rangle M$.

(c) Assume x is a nonzerodivisor on M and y is a nonzerodivisor on M/xM. Prove that $H_2(x,y;M) = 0$.

(d) Under the same assumptions, prove that $H_1(x,y;M) = 0$. Hint: If $xa + yb = 0$, reduce modulo xM to show that $b \in xM$.

(e) Conclude that if x,y is an M-regular sequence, then $\mathrm{Kos}(x,y;M)$ has no positive-degree homology.

(f) Let $A = \mathbb{K}[x,y]/\langle xy\rangle$. Compute $H_2(x,y;A)$ and find a nonzero class in $H_1(x,y;A)$.

(13) **The General Koszul Complex.** Let $\mathbf{x} = x_1,\dots,x_n$.

(a) Write out the Koszul complex explicitly when $n = 3$.

(b) Verify directly when $n = 3$ that $d^2 = 0$.

(c) Show that $H_0(\mathbf{x};M) \cong M/\langle x_1,\dots,x_n\rangle M$.

(d) Prove that permuting $x_1,\dots,x_n$ changes the Koszul complex only up to a natural isomorphism of complexes. Hint: Permuting the basis of F changes wedge products by signs.

(e) Show that $\mathrm{Kos}(x_1,\dots,x_n;M)$ is naturally isomorphic to the tensor product of the one-element complexes

$$\mathrm{Kos}(x_1;R) \otimes_R \cdots \otimes_R \mathrm{Kos}(x_n;R) \otimes_R M.$$

(14) **Koszul Acyclicity for Regular Sequences.** The goal of this exercise is to prove the following theorem.

Theorem 4 (Koszul Acyclicity)**.** *If $x_1,\dots,x_n$ is an M-regular sequence, then*

$$H_i(x_1,\dots,x_n;M) = 0 \qquad \textit{for all } i > 0,$$

and

$$H_0(x_1,\dots,x_n;M) \cong M/\langle x_1,\dots,x_n\rangle M.$$

Let $\mathbf{x}' = x_1,\dots,x_{n-1}$ and write $K' = \mathrm{Kos}(\mathbf{x}';M)$.

(a) Explain why $\mathrm{Kos}(\mathbf{x};M)$ is obtained from K' by tensoring with the two-term complex $\mathrm{Kos}(x_n;R)$.

(b) Equivalently, show that $\mathrm{Kos}(\mathbf{x};M)$ is the mapping cone of the chain map

$$K' \xrightarrow{x_n} K'$$

given by multiplication by x_n, up to a harmless sign convention.

(c) Use the long exact sequence in homology of a mapping cone to obtain exact sequences

$$H_i(K') \xrightarrow{x_n} H_i(K') \longrightarrow H_i(\mathbf{x}; M) \longrightarrow H_{i-1}(K') \xrightarrow{x_n} H_{i-1}(K').$$

(d) Prove the acyclicity theorem by induction on n.

(e) Apply the theorem to compute the Koszul homology of $x_1, \ldots, x_n$ on $\mathbb{K}[x_1, \ldots, x_n]$.

The Koszul complex also computes depth. Let $(R, \mathfrak{m}, \mathbb{K})$ be a Noetherian local ring, let M be a finitely generated R-module, and suppose $\mathfrak{m} = \langle x_1, \ldots, x_n \rangle$. Then the Koszul homology modules $H_i(x_1, \ldots, x_n; M)$ measure how far the chosen generators are from being an M-regular sequence.

Theorem 5 (Depth from Koszul Homology)**.** *Let $(R, \mathfrak{m})$ be a Noetherian local ring, let M be a finitely generated nonzero R-module, and let $x_1, \ldots, x_n$ generate $\mathfrak{m}$. Then*

$$\operatorname{depth}_R(M) = n - \max\{i \mid H_i(x_1, \ldots, x_n; M) \neq 0\}.$$

In particular, the number on the right is independent of the chosen generating set for $\mathfrak{m}$.

This theorem should be read as follows: the later the Koszul homology survives, the smaller the depth is. If all positive Koszul homology vanishes, then the generators themselves form a regular sequence and the depth is as large as possible.

---

(15) **Computing Depth with Koszul Homology.** Let $(R, \mathfrak{m}, \mathbb{K})$ be Noetherian local and suppose $\mathfrak{m} = \langle x_1, \ldots, x_n \rangle$.

(a) Compute $\operatorname{Kos}(x_1, \ldots, x_n; \mathbb{K})$. What are the differentials?

(b) Show that

$$H_i(x_1, \ldots, x_n; \mathbb{K}) \cong \bigwedge^i \mathbb{K}^n.$$

Use the depth theorem to recover $\operatorname{depth}_R(\mathbb{K}) = 0$.

(c) Let $R = \mathbb{K}[x_1, \ldots, x_n]_{\langle x_1, \ldots, x_n \rangle}$. Use Koszul acyclicity to compute $\operatorname{depth}(R)$.

(d) Let $A = \mathbb{K}[x, y]_{\langle x, y \rangle} / \langle xy \rangle$. Show that $H_2(x, y; A) = 0$ and that the class of $(0, \overline{x}) \in A^2$ is a nonzero class in $H_1(x, y; A)$. Use the depth theorem to compute $\operatorname{depth}(A)$.

(e) Let $B = \mathbb{K}[x, y]_{\langle x, y \rangle} / \langle x^2, xy \rangle$. Show that $\overline{x} \in B$ determines a nonzero class in $H_2(x, y; B)$. Use the depth theorem to compute $\operatorname{depth}(B)$.

(16) **Top Koszul Homology and Socles.** Let $(R, \mathfrak{m})$ be local and suppose $\mathfrak{m} = \langle x_1, \ldots, x_n \rangle$.

(a) Show that the top Koszul differential has kernel

$$H_n(x_1, \ldots, x_n; M) \cong (0 :_M \mathfrak{m}).$$

Hint: The top differential sends m to the tuple whose entries are, up to signs, $x_1 m, \ldots, x_n m$.

(b) The module $(0 :_M \mathfrak{m})$ is called the *socle* of M. Compute the socle of $\mathbb{K}[x,y]_{\langle x,y\rangle}/\langle x^2, xy\rangle$.

(c) Explain why nonzero top Koszul homology forces depth zero.

(d) If M has finite length, prove that its socle is nonzero. Deduce that every nonzero finite-length module over a local ring has depth zero.

(17) **A Final Synthesis.** Let $S = \mathbb{K}[x,y,z]_{\langle x,y,z\rangle}$.

(a) Show that x, yz is an S-regular sequence.

(b) Let $A = S/\langle x, yz\rangle$. Find the minimal primes of A and compute $\dim(A)$.

(c) Compute $\operatorname{depth}(A)$ using the theorem that depth drops along regular sequences.

(d) Now compute the same depth using the Koszul complex on a generating set of the maximal ideal of A.

(e) Compare this example with $C = S/\langle xy, xz\rangle$: find a minimal prime of height one over $\langle xy, xz\rangle$ and explain why xy, xz cannot be a regular sequence.